# PRIYANKA2

**prepare a prototype (flow + structure)** for a **Document Management System (DMS)** based on your detailed workflow (Creator → Approver → Issuer → Controlled Access).

-System Flow Chart (process steps)

-Module-wise Functionality (dashboard, upload, approval, issuance, access control)

-Database Design (sample schema)

-User Role Permissions

-Technology suggestion for development

Creator Login

Fill Doc Info (a–g fields)

Upload Word Doc (No header/footer)

Send to Approver (Notify)

Approver reviews doc (with header/footer)

Approve Document Decline (add remark)

Add remarks, Goes back to Creator

select sharing dept.

Send to Issuer (MR)

Issuer reviews doc (header/footer)

Issue Final Copy Decline (add remark)

Add Issue remark Goes back to Creator

Convert to PDF (Final)

Auto Notify All Users

Assign Control Copy Nos

Users View/Print Controlled PDF only

(No download, traceable by User ID)

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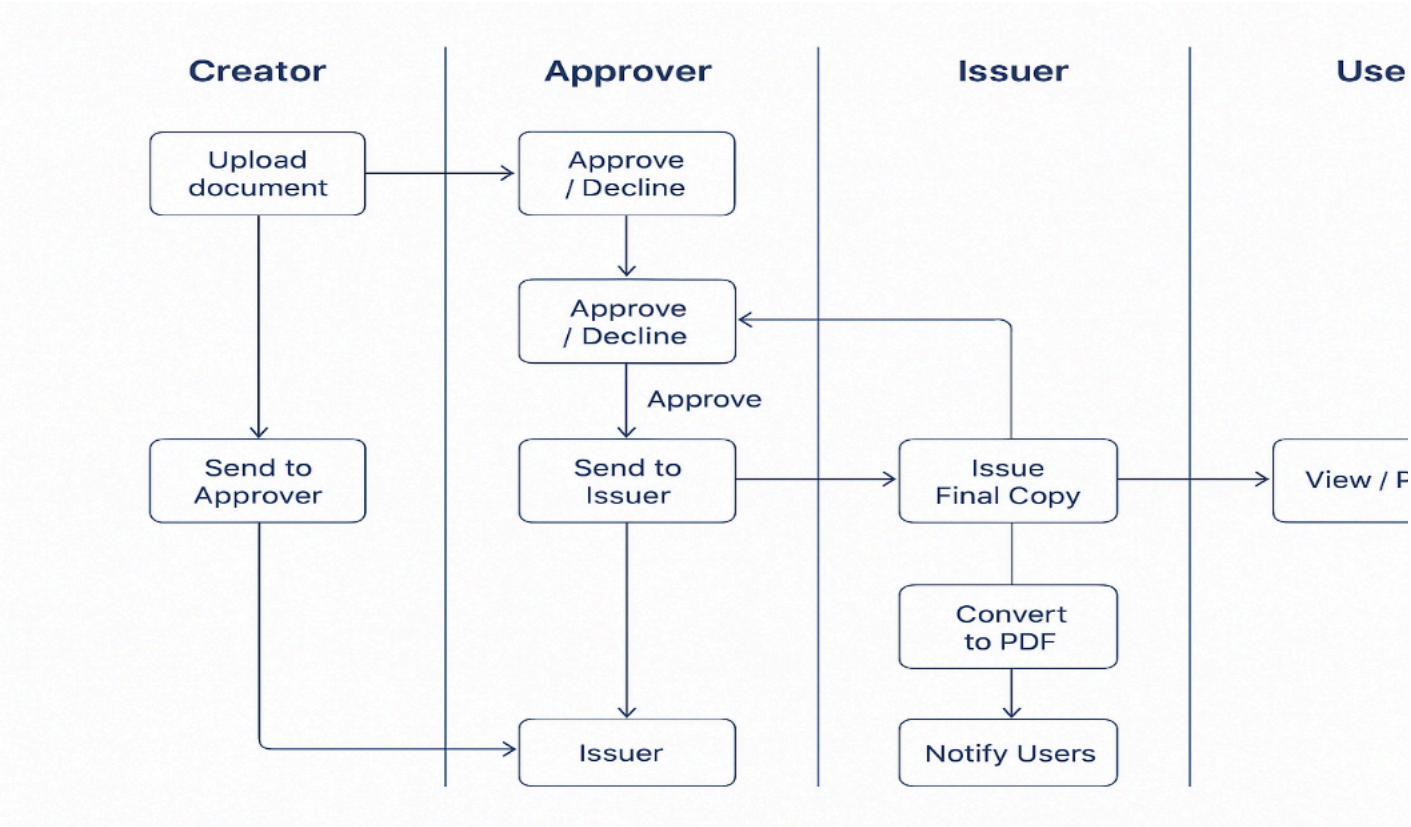
**Responsibility: Module-Wise Functional Prototype Plan**

| Role | Features |
|---|---|
| **Creator** | Create new document entry, upload Word doc, view pending/returned docs, resubmit |
| **Approver** | View pending approvals, open doc (auto header/footer), approve/reject, add remarks, assign departments |
| **Issuer (MR)** | View approved docs, issue/reject, add final remark, track issued documents |
| **User/Recipient** | View issued documents (PDF only), print with control footer |
| **Admin** | Manage users, roles, departments, track logs, export reports |




**There is some conditions:**

-Auto reminders for revision (based on due period)

-Comparison tool (old vs. revised doc)

-Version history tracking

-Export to Excel logs (per user or document)

-QR code footer for authenticity

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About Database table need:

users

| Column | Type | Description |
|---|---|---|
| user_id | INT (PK) | Unique user |
| name | VARCHAR | User name |
| role | ENUM('Creator','Approver','Issuer','User','Admin') | Role |
| department | VARCHAR | Dept name |
| email | VARCHAR | Notification use |

| documents<br>Column | Type | Description |
|---|---|---|
| doc_id | INT (PK) | Unique document |
| doc_name | VARCHAR | SOP/Format name |
| doc_number | VARCHAR | Unique doc code |
| date_of_issue | DATE | Issue date |
| revision_no | INT | Revision count |
| due_period_years | INT | Review reminder |

| documents<br>**Column** | **Type** | **Description** |
|---|---|---|
| prepared_by | INT (FK) | Creator ID |
| reason_revision | TEXT | Revision reason |
| file_path | VARCHAR | Uploaded file |
| status | ENUM('Pending','Approved','Declined','Issued') | Current status |
| approver_id | INT (FK) | Assigned approver |
| issuer_id | INT (FK) | Assigned issuer |

document_log

| **Column** | **Type** | **Description** |
|---|---|---|
| log_id | INT (PK) | Log entry |
| doc_id | INT (FK) | Related document |
| user_id | INT (FK) | Who performed action |
| action | VARCHAR | Created, Approved, Declined, Issued |
| remarks | TEXT | Action remarks |
| date_time | DATETIME | Timestamp |

| **document_access**<br><br>**Column** | **Type** | **Description** |
|---|---|---|
| doc_id | INT (FK) | Document |
| user_id | INT (FK) | Recipient |
| control_copy_no | VARCHAR | Auto serial copy no. |
| view_access | BOOLEAN | Can view PDF |
| print_access | BOOLEAN | Can print PDF |
| print_date | DATETIME | Track printed date |

| **Feature** | **Creator** | **Approver** | **Issuer** | **Recipient** | **Admin** |
|---|---|---|---|---|---|
| Create Doc | ✅ | ❌ | ❌ | ❌ | ✅ |
| Approve Doc | ❌ | ✅ | ❌ | ❌ | ✅ |
| Issue Doc | ❌ | ❌ | ✅ | ❌ | ✅ |
| View PDF | ✅ (Issued only) | ✅ | ✅ | ✅ | ✅ |
| View Previous Revisions | ❌ | ❌ | ✅ (Master copy) | ❌ | ✅ |
| Export Logs | ✅ | ✅ | ✅ | ❌ | ✅ |

| Layer | Technology |
|---|---|
| **Frontend (Web UI)** | React.js / Next.js (with Role-based Dashboard) |
| **Backend API** | Node.js (Express) or ASP.NET Core |
| **Database** | PostgreSQL / SQL Server |
| **File Storage** | Local folder or AWS S3 / Supabase Storage |
| **Notifications** | Email + In-app popup via Socket.io or Supabase Realtime |
| **Document Conversion** | LibreOffice / Word API → PDF (with watermark + control footer) |
| **Audit Logging** | Automatic via middleware (store all user actions) |